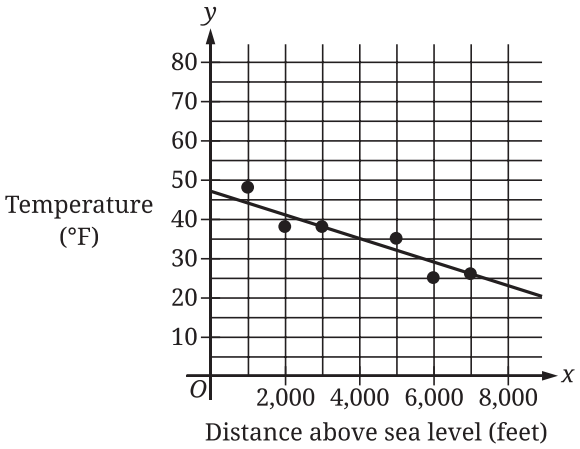
1

The scatterplot shows the temperature, in degrees Fahrenheit (°F), and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown.



At a distance of 4,000 feet above sea level, what is the temperature, in °F, predicted by the line of best fit?

A) 47

B) 35

C) 25

D) 0

2

Rectangle P has an area of 72 square inches. If a rectangle with an area of 20 square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

A) 92

B) 84

C) 80

D) 52

3

$$|p| + 61 = 65$$

Which value is a solution to the given equation?

A) $\frac{65}{61}$

B) 4

C) 126

D) 130

4

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid \$2 for the box of cereal and \$1.90 per pound for the strawberries. If Lorenzo paid a total of \$9.60 for the box of cereal and the strawberries, which of the following equations can be used to find p, the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

A) $1.90p + 2 = 9.60$

B) $1.90p - 2 = 9.60$

C) $1.90 + 2p = 9.60$

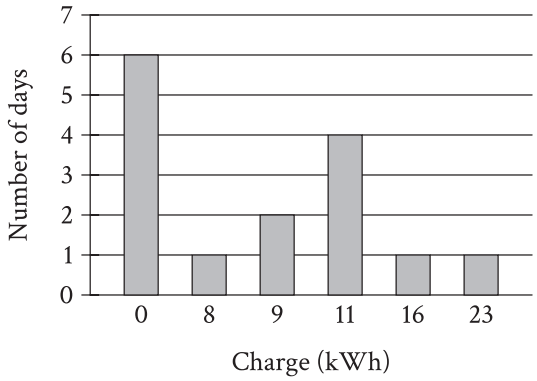
D) $1.90 - 2p = 9.60$

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CONTINUE

5

The bar graph summarizes the charge, in kilowatt-hours (kWh), a battery received each day for 15 days.



For how many of these 15 days did the battery receive a charge of 0 kWh?

A) 0

B) 1

C) 4

D) 6

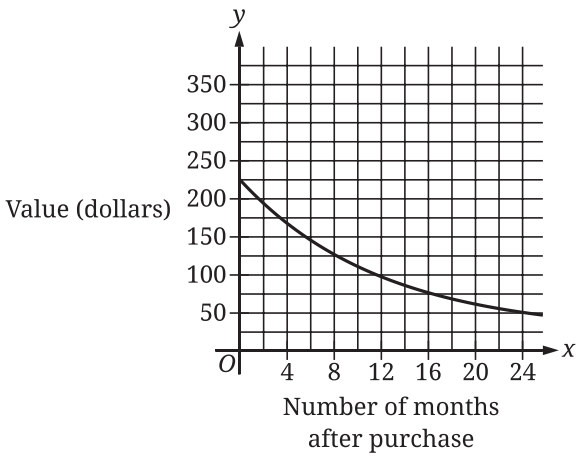
6

A line in the xy-plane has a slope of 9 and passes through the point $(0, -5)$. The equation $y = px + r$ defines the line, where p and r are constants. What is the value of p ?

7

What is an x-coordinate of an x-intercept of the graph of $y = 3(x - 14)(x + 5)(x + 4)$ in the xy-plane?

8



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

A) The estimated value of the tablet was \$225 when it was purchased.

B) The estimated value of the tablet 24 months after it was purchased was \$225.

C) The estimated value of the tablet had decreased by \$225 in the 24 months after it was purchased.

D) The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

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CONTINUE

9

Triangles EFG and JKL are congruent, where E, F, and G correspond to J, K, and L, respectively. The measure of angle E is 45° and the measure of angle F is 20°. What is the measure of angle J ?

A) 20°

B) 45°

C) 135°

D) 160°

10

The function f is defined by $f(x) = \frac{1}{2}(x + 6)$. What is the value of $f(4)$?

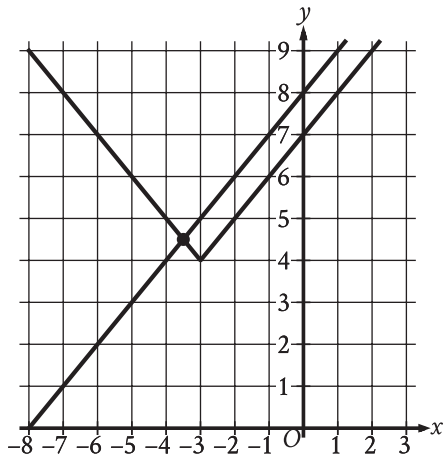
A) 20

B) 12

C) 10

D) 5

11



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

A) $(0, 8)$

B) $\left(\frac{7}{2}, \frac{9}{2}\right)$

C) $\left(-\frac{7}{2}, \frac{9}{2}\right)$

D) $(-3, 4)$

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CONTINUE

12

$$y = 6x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

A) $y = 2(6x) + 3$

B) $y = 2(6x + 3)$

C) $2(y) = 2(6x) + 3$

D) $2(y) = 2(6x + 3)$

13

If $\frac{6}{7}p + 18 = 54$, what is the value of $7p$?

14

$$y = 9x + 12$$
$$x + 7y = 20$$

The solution to the given system of equations is (x, y). What is the value of y ?

15

A circle in the xy-plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

A) The center is at $(13, k)$ and the radius is 8.

B) The center is at $(k, 13)$ and the radius is 8.

C) The center is at $(k, 13)$ and the radius is 64.

D) The center is at $(13, k)$ and the radius is 64.

16

The function f is defined by $f(x) = |x - 4x|$. What value of a satisfies $f(5) - f(a) = -15$?

A) -20

B) 5

C) 10

D) 45

17

For the exponential function f, the value of $f(0)$ is c, where c is a constant. Of the following equations that define the function f, which equation shows the value of c as the coefficient or the base?

A) $f(x) = 22(1.5)^{x+1}$

B) $f(x) = 33(1.5)^{x}$

C) $f(x) = 49.5(1.5)^{x-1}$

D) $f(x) = 74.25(1.5)^{x-2}$

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CONTINUE

18

The function $f(t) = 40{,}000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

A) 2
B) 790
C) 1,580
D) 40,000

19

$$\frac{12}{n} - \frac{2}{t} = -\frac{2}{w}$$

The given equation relates the variables n, t, and w, where $n > 0$, $t > 0$, and $w > t$. Which expression is equivalent to n ?

A) $12tw$

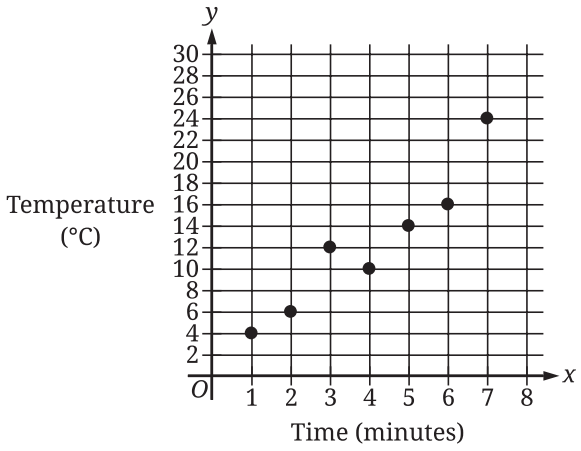
B) $6(t - w)$

C) $\frac{w - t}{6tw}$

D) $\frac{6tw}{w - t}$

20

During a study, the temperature, in degrees Celsius (°C), of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature y, in °C, of the air in the chamber x minutes after the start of the study.



What was the average rate of change, in °C per minute, of the recorded temperature of the air in the chamber from $x = 5$ to $x = 7$?

21

In August, a car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. In August and September, the car dealer completed 363 sales. How many sales did the car dealer complete in September?

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CONTINUE

22

Points Q and R lie on a circle with center P. The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\overline{QR}$?

A) $13\sqrt{2}$

B) 13

C) $9\sqrt{2}$

D) 9

23

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x. The product of 12 and the fourth odd integer is at most 26 less than the sum of the first and third odd integers. Which inequality represents this situation?

A) $12(x + 6) \leq x + (x + 4) - 26$

B) $12(x + 6) \geq 26 - (x + (x + 4))$

C) $12(x + 4) \leq x + (x + 3) - 26$

D) $12(x + 4) \geq 26 - (x + (x + 3))$

24

x	y
$-2s$	24
$-s$	21
s	15

The table shows three values of x and their corresponding values of y, where s is a constant. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A) $sx + 3y = 18s$

B) $3x + sy = 18s$

C) $3x + sy = 18$

D) $sx + 3y = 18$

25

Which of the following expressions is equivalent to $(\sin 24°)(\cos 66°) + (\cos 24°)(\sin 66°)$?

A) $2(\cos 66°)(\sin 24°)$

B) $2(\cos 66°) + 2(\cos 24°)$

C) $(\cos 66°)^2 + (\cos 24°)^2$

D) $(\cos 66°)^2 + (\sin 24°)^2$

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CONTINUE

26

The cost of renting a carpet cleaner is $52 for the first day and $26 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

A) $C(d) = 26d + 26$

B) $C(d) = 26d + 52$

C) $C(d) = 52d - 26$

D) $C(d) = 52d + 78$

27

$$f(x) = (x - 2)(x + 15)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

STOP

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1

A total of 165 people contributed to a charity event as either a donor or a volunteer. 130 people contributed as a donor. How many people contributed as a volunteer?

A) 35

B) 130

C) 165

D) 330

2

There are 250 trees in a park. Of these trees, 6% are birch trees. How many birch trees are in the park?

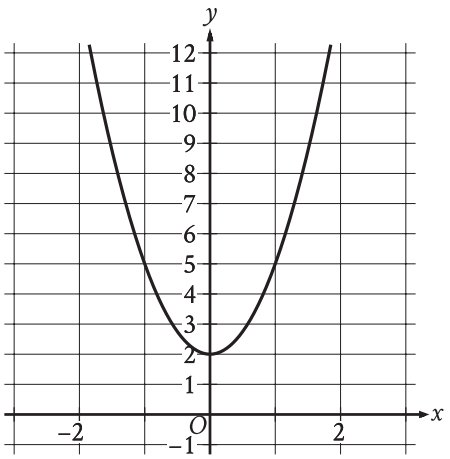
A) 6

B) 15

C) 75

D) 244

3



The graph of the quadratic function $y = f(x)$ is shown. What is the vertex of the graph?

A) $(0, -2)$

B) $(0, -3)$

C) $(0, 2)$

D) $(0, 3)$

4

The number of raccoons in a 131-square-mile area is estimated to be 2,358. What is the estimated population density, in raccoons per square mile, of this area?

A) 18

B) 131

C) 149

D) 2,376

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CONTINUE

5

−11, −9, 26

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a positive number?

A) 0

B) $\frac{1}{3}$

C) $\frac{2}{3}$

D) 1

6

$f(x) = 45x + 600$

The function f gives the monthly fee $f(x)$, in dollars, a facility charges to keep x crates in storage. What is the monthly fee, in dollars, the facility charges to keep 50 crates in storage?

7

The function f is defined by $f(x) = 5\left(\frac{1}{4} - x\right)^2 + \frac{11}{4}$.

What is the value of $f\left(\frac{1}{4}\right)$?

8

If $8x = 6$, what is the value of $72x$?

A) 3

B) 15

C) 54

D) 57

9

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

A) $23x(x^2 + 2x + 9)$

B) $9x(23x^3 + 2x^2 + 1)$

C) $x(23x^2 + 2x + 9)$

D) $34(x^3 + x^2 + x)$

10

Which expression is equivalent to $(9x^3 + 5x + 7) + (6x^3 + 5x^2 - 5)$?

A) $15x^6 + 5x^2 - 5x - 35$

B) $15x^3 + 10x^2 + 2$

C) $15x^6 + 5x^2 + 5x + 2$

D) $15x^3 + 5x^2 + 5x + 2$

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CONTINUE

11

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of 1,120 points. The equation $80S + 90C = 1{,}120$ represents this situation. How many more points is a circle token worth than a square token?

A) 950

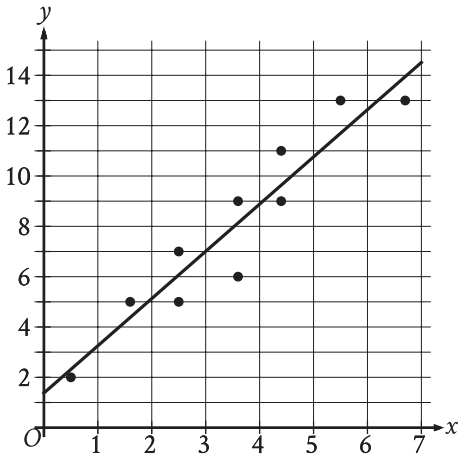
B) 90

C) 80

D) 10

12

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit shown?

A) 0

B) $\frac{1}{2}$

C) 1

D) 2

13

A circle has a radius of 2.1 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

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CONTINUE

14

In triangle XYZ, angle Y is a right angle, point P lies on $\overline{XZ}$, and point Q lies on $\overline{YZ}$ such that $\overline{PQ}$ is parallel to $\overline{XY}$. If the measure of angle XZY is 63°, what is the measure, in degrees, of angle XPQ ?

15

An investment account was opened with an initial value of \$890. The value of the account doubled every 10 years. Which equation represents the value of the account $M(t)$, in dollars, t years after the account was opened?

A) $M(t) = 890\left(\frac{1}{2}\right)^{\frac{t}{10}}$

B) $M(t) = 890\left(\frac{1}{10}\right)^{\frac{t}{2}}$

C) $M(t) = 890(2)^{\frac{t}{10}}$

D) $M(t) = 890(10)^{\frac{t}{2}}$

16

$$y < x$$
$$x < 22$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	y
19	18
20	19
21	20

B)

x	y
19	20
20	21
21	22

C)

x	y
23	22
24	23
25	24

D)

x	y
23	24
24	25
25	26

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CONTINUE

17

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

A) $\frac{h^{10}}{q^{14}}$

B) $\frac{h^3}{q^3}$

C) $h^{10}q^{14}$

D) h^3q^3

18

$$3y = 4x + 17$$
$$-3y = 9x - 23$$

The solution to the given system of equations is (x, y). What is the value of $39x$?

A) -18

B) -6

C) 6

D) 18

19

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 3,364 feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

A) 7.25

B) 14.50

C) 105.13

D) 210.25

20

$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as $\frac{8 - \sqrt{k}}{4}$, where k is a constant. What is the value of k ?

21

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x - 560)°$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)°$. What is the value of w ?

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CONTINUE

22

x	$f(x)$
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For the linear function f, the table shows three values of x and their corresponding values of $f(x)$. If $h(x) = f(x) - 13$, which equation defines h ?

A) $h(x) = 5x - 4$

B) $h(x) = 5x + 7$

C) $h(x) = 5x + 9$

D) $h(x) = 5x + 20$

23

The linear function g is defined by $g(x) = b - 15x$, where b is a constant. If $g(c + 7) = \frac{c}{4}$, where c is a constant, which of the following expressions represents the value of b ?

A) $\frac{15c}{4}$

B) $\frac{19c}{4} + 7$

C) $\frac{61c}{4} + 105$

D) $15c + 105$

24

In triangle XYZ, angle Z is a right angle and the length of $\overline{YZ}$ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

A) 188

B) 168

C) 84

D) 71

25

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy-plane, the graph of the given equation is a circle. What is the length of the circle's radius?

A) $\sqrt{109}$

B) $\sqrt{149}$

C) $\sqrt{167}$

D) $\sqrt{341}$

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CONTINUE

26

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

A) 0.3

B) 16.3

C) 195.8

D) 220.4

27

$$y = -2.5$$

$$y = x^2 + 8x + k$$

In the given system of equations, k is a positive integer constant. The system has no real solutions. What is the least possible value of k ?

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